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Chapter 1

Synthesis of a Self-Healing Polymer

1.1 Results

1.1.1 Hydrolysis of the Polyimide

The method used was based on previous literature publications[1–8] where an alkaline solution is used to hydrolyse polyimide to form a salt intermediate and then an acidification to form carboxlyic acid groups. A NaOH solution (0.2M)[2–4] was initially trialled for the alkaline hydrolysis step at room temperature for 90 mins, however when there was minimal change under Infrared spectrometry (IR), the temperature of the reaction was increased to 120°C. Also, the NaOH concentration was adapted similar to that used in Xue (1986).[7] This comprised of the use of 1 equiv. OH^- for every 1 monomer of the polyimide. This was chosen to ensure a partial hydrolysis, to only expose one carboxylic acid group rather than the total hydrolysis to form a tetrasubstituted benzene ring. However during analysis of this reactio we also observed minimal hydrolysis via IR.

A widely accepted base-catalysed amide hydrolysis route is through a second order pathway, where a hydroxy group (OH^-), from the base, nucleophillically adds to the carbonyl group (C=O) to form a tetrahedral intermediate.[9] This cleaves into two and the amine

is eliminated. In pthalamide structures, a molecule close in character to the polyimide monomer, has been shown to exhibit similar behaviour when in solution.[10]

However, mechanistic changes occur when the reaction contains macromolecules, it has been shown in previous literature that there is a third order dependence on the concentration of OH^-([OH^-]) and follows a mechanism that starts with the nucleophillic addition of the OH^- to form a charged intermediate before undergoing rate-determining cleavage of the C-N bond to form the associated polyamic acid.[2] Moens (1957) proposed that, due the hydrostatic repulsion, the rate of hydrolysis is lowered when a polymeric chain has increasing numbers of negative charges,[11] this would imply that the formation of the monanion intermediate could hinder the reaction.

While multiple literature sources opt to use KOH,[1,5,7,8] hydrolysis has been shown to occur in the presence of NaOH.[2–4] In retrospect, hydrolysis attempts using KOH should have also been approached due to the success of many KOH hydrolysis attempts.

However, we attribute most of the failure in this experiment to the low concentration of NaOH, as sources have stated that a concentration below 0.5M was ineffective.[1] This explains why the testing with 0.04M was also ineffective. Additionally, some of the failure may be caused by the lack of addition of a reagent to maintain constant ionic strength, which has been shown to assist hydrolysis with NaOH.[2]

Spectral Analysis

Structural determination was conducted via Infrared spectroscopy (IR). A comparison was made between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

Characteristic peaks of both the hydrolysed and the control forms are observed around 1500cm^{-1} , corresponding to the C=C bond in the central benzene ring[1,5] along with the 1235 cm^{-1} aromatic ether stretch[5] both which do not change throughout the hydrolysis.

Sample	Base	Conc. /M	A_{1500}	A_{1720}	A_{1720}/A_{1500}	Rel. imide content
90 min, 25°C	NaOH	0.2	0.2243	0.2304	1.0273	0.9424
20 min, 120°C	NaOH	0.03	0.1062	0.1167	1.0990	1.0082
40 min, 120°C	NaOH	0.03	0.1608	0.1735	1.0789	0.9898
Kapton	—	—	0.1125	0.1226	1.0900	1.0000
PAA	—	—	0.8508	0.1735	0.2040	0.1871
Wu (2017)	KOH	0	0.0513	0.0587	1.1440	1.0000
Wu (2017)	KOH	0.1	0.0504	0.0573	1.1375	0.9943
Wu (2017)	KOH	0.5	0.0511	0.0568	1.1096	0.9699
Wu (2017)	KOH	1	0.0587	0.0617	1.0519	0.9194
Wu (2017)	KOH	5	0.0763	0.0072	0.0946	0.0827

Table 1.1: ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[1]

Also observed is the C=O asymmetric stretching mode at around 1710 cm^{-1},[1,3,5–7] the peak around 1775 cm^{-1} as the symmetric C=O stretch[5–7] and the C-N-C imide ring stretch at 1370 cm^{-1},[5,6] all of which decrease in magnitude during hydrolysis.[1,6]

One literature source[12] states that the symmetric C=O stretch is at 1775 cm^{-1} and the asymmetric is at 1720 cm^{-1} however, due to the disparity between this and the other sources, we disregarded this result.

Some sources estimate their degree of hydrolysis with a comparison of the ratio of intensities of the C=O : C=C (1720:1500 cm^{-1}) peaks,[1,7] these are summarised in the table 1.1. In this report, the symmetric C=O peak was observed slightly lower at around 1710 cm^{-1}.

As in the Wu 2017 paper,[1] this report assumes the Kapton sample to have 100% imide content on the surface, hence finding that the polyamic acid sample contains 18.71% surface imide content. Also observed was the decrease in imide content relative to both an increase in NaOH concentration and time. This work found that the best hydrolysis attempt was the 90 minute, 0.2M hydrolysis at room temperature, giving a total imide surface content of 94.24%, meaning that 5.76% of the imide rings were cleaved.

1.1.2 Spectrophotometry

Due to the limitations of only using one method of analysis of the degree of hydrolysis of Kapton, an additional method was sought out. Other traditional methods of determination such as nuclear magnetic resonance (NMR) rely on solubilty of samples, this is a property which, by design, Kapton does not have. Therefore, a spectrophotometric test was proposed.

Other spectrophotometric methods of detection of carboxylic acids involve fluorescent tagging. These involve the attachment of a fluorescent label to the carboxylic acid group, producing a distinct peak in a UV-vis spectrum. Attachment reagents involved are varied, including halogenoalkyl, hydrazine, amine, alcohols and sulfonate groups. However, most of these options require activation of the COOH group before binding of the fluorescent marker. We aimed to avoid any pre-activation reagents as this would lower the activation of the unwanted intramolecular amide formation, thereby reforming an imide ring.[13]

Therefore, a method proposed by Johar (1972) involving the use of Eosin Y (EY) and Methylene blue (MB) for determination of carboxylic acids was used.[14]

Dyes were mixed as according to the concentrations stated by Johar (1972). However, is was apparent that ther concentration was too high, therefore additional dilutions of the dye mixtures were made. Initally tested were the interactions between the dyes and water, we observed a peak at (515 ish) confirmed to be corresponding to EY[15–17] and a peak at (650 ish) correspoding to MB.[18–21] The shoulder on the EY peak is formed through the dimerisation of the molecule due to the high concentration.[16]

An example of a postive test would occur with acetic acid, a simple carboxylic acid. In the spectophotometer, the peak corresponding to the EY was destroyed. This report proposes that this destruction is due to the use of glacial (the most concentrated an acid can be) acetic acid. In high concentrations, acetic acid can to convert the EY into a precursor (dihydrofluoran)[22] which causes the observed blue shift in the peak[23] along

with the reduction in magnitude.

Next assessed was the interaction between dye mixture and polyamic acid (PAA) dissolved in dimethyl sulfoxide (DMSO). This gave two peaks, one in the EY region at (515ish) and one in the MB region at (650ish). The interaction with the MB was very similar to that of the acetic acid which was a promising result. However, in the EY region, the large peak was not similar to that of acetic acid. We propose this effect was caused by EY catalysing amide formation, hence forming imide rings and removing the carboxylic acids. This implies that the peak at (515ish) corresponds to the solvent (DMSO) rather than an interaction with the PAA.

To test this hypothesis, DMSO was mixed with the dyes and a peak at (515ish) and (650ish) was observed, which we confirm to be EY[16] (maybe aslo cite aroteinvestigating) and MB[18,20](maybe also cite tchaikovskaya2023solvent). As a final validation, we checked that the interaction was not simply pH based by testing HCl, this was chosen as an acid that does not contain carboxyl groups and therefore should react differently to the acetic acid. The absorbance spectrum showed one peak at (450), which we believe is due to the ....

Due to the difference in absorbance for the acetic acid and the PAA and the catalytic effects of EY on imide reformation, we determined that the GS Johar method is not appropriate for a determination of hydrolysis of Kapton.

1.1.3 Cross-linking of the Polyimide

Cross-linking attempts follow a route where ethylene carbonate (EC) is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them together with a "cross-link". This was attempted with three different salts (Et_4NBr,$ZnCl_2$, $FeCl_3$) as catalysts.

During the cross-linking reaction with Et_4NBr, an orange-brown suspension was formed

that when washed, filtered and freeze dried yielded a pale orange powder. With $ZnCl_2$, the liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, yielded a dark orange/brown gel, which formed a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

During a cross-linking with ethylene carbonate, CO_2 is released. This was a concern as the monowave reactor has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, an artificial limit of 5 bar was set. The Ideal Gas law was used to give the amount of EC to use to ensure this limit was held.

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

P = 500 kPa, V = 10 mL, R = 8.314 J K^{-}1 mol^{-}1, T = 423K

$$n = \frac{500 \times 10 \times 10^{-3}}{8.314 \times 423} = 0.00142mol$$

Since the CO_2 is generated from the EC, there are same number of moles for the CO_2 and for the EC. This motivates the use of $mass = Mr \times n$, where Mr = 88, n = 0.00142

$$Mr \times n = 88 \times 0.00142 = 0.12496g$$

Detailed mechanistic studies of the reaction between a carboxylic acid and ethylene carbonate in the presence of a quaternary ammonium salt, such as Et_4NBr, have been performed in other literature.[24–26] Yoshino (1977) states that the mechanism begins with a nucleophillic ativation by the anion of the catalytic salt on a methylene group of the EC, this cleaves the hydrogen from the alcohol group (OH) of the carboxylic acid which can

now nucleophillically substitute the halide to form the ester. Fagerburg (1984) disagreed with this mechanism as they proposed that the rate of the reaction was be dependent on the anion of the catalytic salt. However they did not find this experimentally. They observed during a reaction of the catalytic salt and EC alone, the activation energy of the EC decomposition was much higher than any esterification attempts . They therefore proposed a mechanism involving the attack of a carboxylate species on the methylene carbon of the EC to decompose the ring and form the ester product. Given the depth of Fagerburg's work, we consider this mechanism to be more robust than the one proposed by Yoshino.

The change of catalyst to a lewis acid ($ZnCl_2, FeCl_3$) was to encourage a direct activation of the carbonyl group on the EC, the mechanism for this proposed step was as follows: The lewis acid activates the carbonyl on the EC by forming a "complex". This allows the alkoxide to attack the methylene group of the EC and form the hydroxyethyl ester.[27] Von Seggern (2020) however reported, the use of zinc based catalysts was found to be ineffective in this reaction.[27]

Spectral analysis

Spectral analysis was performed mainly with IR, but also with ^{1}H NMR and DOSY.

From the IR data, the crosslinking with Et_4NBr was deemed to be unsuccessful as the product spectra the key peaks of the C-O bond corresponding to an ester or any alcohol formation were not observed. Alcohol peaks would be likely as crosslinking attempts are unlikely to be 100% successful, hence the single ester product would also be formed. However, neither of these were present and since the product was insoluble, no NMR could be obtained. This IR spectrum was compared with the spectrum of Kapton (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. Key spectral peaks confirming the lack of crosslinking contain the C-N-C peak, commonly found at 1370 cm^{-1}, however in our sample we observed this

peak at 1391 cm^{-1}. Other signs of imidisation include the increase in the magnitude of the C=O stretch at 1715 cm^{-1}. These peaks would indicate the imide ring reformation as opposed to the esterification product.

From the IR data of the $ZnCl_2$ and the $FeCl_3$ attempts, we observe two new peaks at around 1800 cm^{-1} which are not in either the Kapton or the PAA samples. This shows the presence of EC in the final sample as C=O bonds in liquid EC[28] are found in this regon. Since these samples were soluble, proton nuclear magnetic resonance (^{1}H NMR) was available. A new peak was observed at 4.43ppm in the ^{1}H NMR spectrum, however, when adding 2 μL of EC in a "spike test", the peak increased in size confirming that the new peak was starting material (EC). Other peaks in the $ZnCl_2$ and $FeCl_3$ samples are indicative of unreacted polyamic acid. After increasing the temperature of the $ZnCl_2$ catalysed sample, the peaks positions show the intramolecular cyclisation of the imide ring with the C-N-C stretch at 1370cm^{-1}.

DOSY data was collected for the $ZnCl_2$ sample ....

1.1.4 Catalytic Imidisation of polyimide

During the attempted crosslinking reactions, the imide ring formation was observed at a lower temperature than previously seen in the literature. This motivated attempts to recreate this with only polyamic acid (PAA) and $ZnCl_2$.

Without the catalyst, material got slightly darker after heating for an hour. This was attributed to some minor solvent evaporation and therefore, substrate becoming more concentrated. A similar colouration after heating the sample containing catalyst was observed, however, there was also material suspended. This was proposed as indicative of a partially successful attempt.

An assessment of the success of the imidisation can be made by comparing the A_{1720} to the A_{1500} in the same manner as previous analysis.[1] In table 1.2, again the Kapton sheet was

Sample	ν_{1500} /cm^{-1}	A_{1500}	ν_{1720} /cm^{-1}	A_{1720}	A_{1720}/A_{1500}	Rel. imide content
PAA	1495.02	0.8508	1709.18	0.1735	0.2040	0.1871
No_cat	1495.02	0.4209	1717.34	0.1375	0.3268	0.2998
With_cat	1497.06	0.0727	1721.42	0.0443	0.6084	0.5581
Kapton	1497.06	0.1125	1709.18	0.1226	1.0900	1.0000

Table 1.2: ATR-FTIR peak positions, absorbances, and relative imide surface content

assumed to have 100% surface imide content and others were calculated relative to this. This report finds that the imide content of the polyamic acid heated without the catalyst had 29.98% imide content as compared to the baseline value from the unheated PAA at 18.71%. However, the sample with $ZnCl_2$ catalyst presents as successful as 55.81% imide content is observed. Also observed was a shift in peak position in the 1500 peak when imide content is over 50%. To confirm the degree of imidisation in the future, a pure sample could be obtained through a recrystallisation from DMSO.

1.2 Experimental

1.2.1 Equipment

A MonoWave reactor (MonoWave 50 Closed Reaction Vessel; ANTON PAAR) was used for the majority of experiments covered in this report. This reaction vessel contains integrated temperature and pressure control and a magnetic stir element. This allowed the user to perform reactions more quickly as samples could be heated above their ambient pressure boiling points. The MonoWave reactor required a specific 10mL tube that was sealed with a rubber cap and a PTFE septum, to ensure agitation, an oval-shaped magnetic stir bar was included.

Dye Mix	Parts water	Parts Eosin Y (0.5%)	Parts Methylene Blue (0.1%)
Mix 1	10	2	1
Mix 2	5	2	1
Mix 3	0	2	1

Table 1.3: Dye mixtures for examination in a spectrophotometer with dilution by number of parts water. Here a "part" is defined as 1.5mL.

1.2.2 The Hydrolysis of a Polyimide

Polyoxydiphenylene mellitimide (Kapton) film (0.18g, thickness: x) was cut into pieces of approx. $3mm^2$ and placed into a beaker. This was stirred at room temperature with a solution of NaOH in DMSO (0.2M, 22ml). There was no observable difference and an Infrared (IR) spectrum was taken.

Another sample of Kapton film was cut into approx. $3mm^2$ pieces and placed into a MonoWave reaction tube along with a solution of NaOH (vol, 0.03M). This tube was placed into a MonoWave reactor that was set to ramp for 20 minutes and hold for 20 minutes at 100°C. There was no observable difference, so a sample was taken and the remaining mixture was placed onto the MonoWave under the same conditions. The mixture was filtered and thoroughly washed with water before IR was collected.

1.2.3 Spectrophotometry as a Determination Method for Hydrolysis

The preparation of samples for the spectrophotometric analysis was as follows:

3 mL 0.5% Eosin Y solution was added to 1.5mL 0.1% Methylene Blue solution. This solution was diluted with 10 parts water and 5 parts water (where a "part" is 1.5mL) to give three dye mixes (summarised in 1.3)

Test compound (2.7mL) was added to a sample vial along with one of the dye mixes (0.3 mL) to form samples summarised in 1.4. These were inverted to ensure mixing before

Sample number	Test compound	Dye mix
S01	Dimethyl Sulfoxide	Dye 1
S02	Polyamic Acid solution	Dye 1
S03	Water	Dye 1
S04	Acetic Acid	Dye 1
S05	Hydrochloric Acid	Dye 1
S06	Dimethyl Sulfoxide	Dye 2
S07	Polyamic Acid solution	Dye 2
S08	Water	Dye 2
S09	Acetic Acid	Dye 2
S10	Hydrochloric Acid	Dye 2
S11	Dimethyl Sulfoxide	Dye 3
S12	Polyamic Acid Solution	Dye 3
S13	Water	Dye 3
S14	Acetic Acid	Dye 3
S15	Hydrochloric Acid	Dye 3

Table 1.4: Samples for testing in a spectrophotometer using dye mixture summarised in Table 1.3.

being transferred to a plastic cuvette and placed in a spectrophotometer (DETAILS) to test between 250 and 750nm.

1.2.4 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of polyamic acid (PAA) via ethylene carbonate (EC) using a catalytic salt as outlined in previously published procedures.

A mixture of polyamic acid (0.51mL, conc) was placed into a MonoWave tube with ethylene carbonate (0.08g) and a catalytic salt. These were placed into the MonoWave to ramp for 20 minutes and hold at the reaction temperature for an hour. The products were washed with water and filtered before being freeze dried and examined on the spectrometers. The reagent amounts and processing methods are summarised in table 1.5.

Catalytic Salt	Mass Salt /g	Reaction Temp. /°C	Analytical Method
Et_4NBr	0.08	150	IR
$FeCl_3$	0.02	100	IR, NMR
$ZnCl_2$	0.03	100	IR, NMR, DOSY

Table 1.5: Catalysts, reaction temperatures and analytical methods used in crosslinking attempts of polyamic acid with ethylene carbonate.

Tetraethylammonium Bromide (Et_4NBr) Crosslinking Spectral Data

A orange powder (grams,moles,yield) was isolated, ν max/cm^{-1}(solid); 1715, 1601, 1501, 1391, 1307, 1252, 1167, 1116, 885, 815, 800, 724 634 606, 567, 516, 428

Iron Chloride ($FeCl_3$) Crosslinking Spectral Data

A white gum (grams,moles,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid); 2931, 1805, 1774, 1721, 1609, 1497, 1403, 1372, 1309, 1256, 1163, 1114, 1075, 987, 822, 759, 724, 679, 606, 563, 479

Zinc Chloride ($ZnCl_2$) Crosslinking at 100°C Spectral Data

A grey gum (g,mol,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid); 3263, 3112, 1679, 1603, 1583, 1530, 1493, 1403, 1367, 1316, 1263, 1187, 1163, 1081, 1036, 1008, 836, 722, 706, 630, 600, 559, 534, 502, 404

Zinc Chloride ($ZnCl_2$) Crosslinking at 110°C Spectral Data

Brittle brown pieces (g,mol,yield) were isolated, ν max/cm^{-1}(solid); 3043, 1776, 1715, 1606, 1538, 1497, 1371, 1307, 1238, 1165, 1114, 1012, 879, 820, 724, 632, 606, 565, 514, 430

1.2.5 Catalytic Imidisation of Polyamic Acid

Polyamic acid (0.51mL), $ZnCl_2$ (0.02g) and a magnetic stirbar were added to a MonoWave tube. This was placed in the MonoWave at 110°C to ramp for 20 minutes and hold at 150°for 60 minutes. A control experiment without the $ZnCl_2$ was also placed under the same conditions. The control experiment gave a clear dark brown viscous liquid which, when dried, gave a gum-like pale orange solid. The test with the $ZnCl_2$ gave a similar dark brown liquid, however this was slightly cloudy and, when processed, gave an orange powder.

Control Spectral Data

IR(solid): vals, vals cm^{-1}

Catalytic Imidisation Spectral Data

IR(solid): vals, vals cm^{-1}

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